

Adaptive Inlier-driven Re-ranking for Efficient Visual Place Recognition

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Abstract

Visual Place Recognition (VPR) is a critical task for applications such as autonomous driving and absolute localization. However, current VPR methods suffer from two fundamental limitations: they do not provide uncertainty estimates on predictions, and they exhibit a significant trade-off between localization accuracy and computational efficiency when applying refinement through image matching. This work addresses these aspects through: (i) an empirical analysis of popular VPR and image matching methods to understand the performance-efficiency trade-off; (ii) investigation of the correlation between retrieval correctness and the number of geometric inliers; (iii) development of an adaptive re-ranking strategy based on logistic regression that predicts query difficulty through the number of inliers with the top-1 retrieved image. The proposed approach dynamically classifies queries as "easy" or "hard" and selectively applies expensive geometric verification only when necessary, achieving significant computational savings (up to 78%) while maintaining high localization accuracy. Results demonstrate robustness across heterogeneous datasets and extreme illumination conditions, validating inlier count as a universal confidence signal for VPR.

1. Introduction

Visual Place Recognition (**VPR**) is the task of recognizing the geographical location where an image was acquired based solely on its visual content. This problem is of fundamental importance for a wide range of real-world applications, including autonomous driving, augmented reality, mobile robotics, absolute pose estimation, and simultaneous localization and mapping.

The VPR problem is typically formulated as an image retrieval task: given a query image, a compact description (**descriptor**) is computed and compared against a database of descriptors from images with known geographical coordinates. Through a similarity search algorithm, most commonly implemented with K-Nearest Neighbors (KNN), the

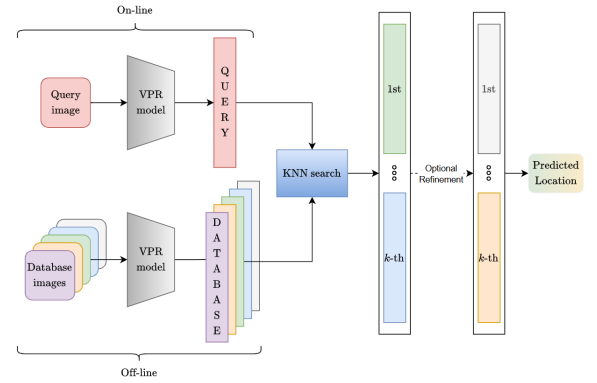


Figure 1. Visual Place Recognition pipeline.

top- K most similar images from a database which were previously run through a VPR model are retrieved. At this point in the pipeline an optional refinement can be achieved using image matching models to re-rank the predictions. The predicted location is then determined by selecting the most similar database descriptor or using some kind of geographical distance metric.

While significant progress has been made in recent years, current VPR methods suffer from critical limitations. One issue is that of the performance-efficiency trade-off in the VPR pipeline: refining retrieval results through local feature matching and re-ranking can significantly improve localization accuracy, but at considerable computational costs.

This project addresses this aspect through: (i) empirical analysis of popular VPR and image matching methods to understand the performance-efficiency trade-off; (ii) investigation of the correlation between queries' correctness using only the retrieval part and the number of inliers within the first retrieved image; (iii) development of an extension focused on improving the computational efficiency of current VPR systems. The implementation and experimental code are publicly available at https://github.com/MatteoSanna23/Visual_Place_Recognition_Project.

2. Related Works

2.1. Visual Place Recognition Methods

The past decade has witnessed substantial advances in VPRs through the development of deep neural network-based approaches. **NetVLAD** [2] stands as a seminal work in the field, introducing a CNN architecture specifically designed for weakly supervised place recognition. The method incorporates a spatial aggregation layer (VLAD) directly into the network, enabling end-to-end learning of discriminative representations suitable for VPR tasks.

Following this foundational work, numerous variants and improvements have been proposed. **CosPlace** [4] reformulates large-scale VPR by introducing new training methodologies and problem formulations. **MixVPR** [1] proposes feature mixing to enhance the robustness of learned representations. More recently, **MegaLoc** [3] presents a unified approach that combines the strengths of multiple methods, advancing state-of-the-art performance.

Importantly, recent work [5] has consolidated the field through comprehensive benchmarking studies, enabling consistent and standardized evaluation of VPR methods using **Recall@N** as the standard evaluation metric. This metric measures the percentage of queries for which at least one of the top- N retrieved database images falls within a geographical distance threshold τ (typically 25 meters) from the ground-truth location.

2.2. Image Matching and Refinement

A crucial component of the VPR pipeline is the optional re-ranking step, which aims to refine initial retrieval results using local feature matching techniques. **SuperGlue** [9] is a learning-based feature matching method that leverages graph neural networks to obtain robust correspondences between local features. **LoFTR** [10] introduces a detector-free approach based on transformers for global local feature matching. More recently, the combination of **SuperPoint** [6] with **LightGlue** [8] provides an efficient solution for local feature matching with reduced computational burden.

Re-ranking through image matching typically relies on the number of inliers (correspondences that survive geometric post-processing via **RANSAC** [7]). However, unconditional application of re-ranking incurs significant computational costs, motivating research into adaptive re-ranking strategies that apply refinement only when necessary.

3. Baseline

3.1. Baseline analysis

To establish a comprehensive understanding of the Visual Place Recognition pipeline and the performance-efficiency trade-off, we conduct a baseline analysis using four VPR methods: **NetVLAD**, **CosPlace**, **MegaLoc**, and **MixVPR**.

For each method, we evaluate the performance of: (i) the retrieval stage alone, and (ii) the retrieval stage combined with three popular image matching methods for re-ranking: **LoFTR**, **SuperGlue**, and **SuperPoint+LightGlue (SP+LG)**.

The evaluation is conducted on four test datasets: **Tokyo-XS** (Tokyo 24/7 dataset subset), **SF-XS** (San Francisco dataset subset), **SVOX Sun** (cross-domain subset), and **SVOX Night** (cross-domain night subset). The primary evaluation metric is **Recall@N** at different thresholds ($N = 1, 5, 10, 20$).

We employ KNN search with $K = 20$ and measure both the L2 and dot product distance to understand their impact on retrieval performance. For the image matching methods, input images are resized to 512×512 pixels. Additionally, we measure the average computational time per query in seconds to quantify the efficiency cost of applying refinement steps.

The results are summarized in Table 1, which presents **R@1**, **R@5**, and **R@10** for each method and dataset combination, along with the average computation time required per query. The first row for each VPR method shows the best **R@20** for each one.

Key observations from baseline analysis:

0) *Difference between distance metric:* Due to normalization of the data by the VPR methods, both distances produce the same results so, from this point on, all of the experiments have utilized the L2 distance.

1) *Performance improvement through re-ranking:* Re-ranking through image matching consistently improves **Recall@1** performance across almost all methods and datasets. The improvements are particularly pronounced for weaker retrieval methods (**NetVLAD**, **CosPlace**), however an already powerful model like **MegaLoc** shows diminishing returns when paired with an image matching model.

2) *Method-specific trade-offs:* **SuperGlue** generally achieves the best computational efficiency (≈ 1.5 - 1.8 seconds per query), while **LoFTR** and **SP+LG** are more computationally expensive (≈ 3 - 4 seconds). However, **LoFTR** often yields slightly better performance improvements in challenging scenarios (e.g., **SVOX Night**).

3) *Dataset-dependent behavior:* The cross-domain **SVOX** datasets show significant performance variations. Particularly, **SVOX Night** presents a challenging scenario where even strong methods like **MegaLoc** benefit from re-ranking, suggesting that domain shift introduces uncertainty that re-ranking can help address.

4) *Necessity of adaptive strategies:* For already high-performing methods like **MegaLoc** (**Recall@1** on **Tokyo-XS**), the marginal improvement from re-ranking (≈ 0 - 3%) may not justify the ≈ 4 second computational overhead. This motivates the development of adaptive re-ranking strategies that selectively apply expensive refine-

Method	tokyo-xs	sf-xs	svox-sun	svox-night	avg Time/ Query (s)
	R@1/R@5/R@10	R@1/R@5/R@10	R@1/R@5/R@10	R@1/R@5/R@10	
NetVlad (R@20)	78.7	56.2	65.8	29.6	
NetVlad	49.8/62.5/70.5	27.2/43.8/50.3	35.4/52.7/58.8	8.0/17.4/23.1	-
NetVlad + LoFTR	68.3/72.7/73.7	53.6/55.3/55.8	61.7/63.1/64.8	25.5/28.1/28.6	≈ 3.42
NetVlad + SuperGlue	69.5/72.7/74.9	53.4/55.3/55.8	60.4/62.9/64.2	24.5/26.6/28.1	≈ 1.49
NetVlad + SP+LG	68.3/72.1/73.7	53.2/55.4/55.8	61.4/63.7/64.5	25.4/27.1/28.4	≈ 3.28
CosPlace (R@20)	85.1	76.9	94.9	80.2	
CosPlace	55.6/74.3/79.7	51.5/66.3/71.8	78.5/89.7/93.1	50.7/67.4/74.8	-
CosPlace + LoFTR	76.5/80.3/81.9	73.8/75.9/76.4	91.7/93.7/94.4	73.6/77.6/79.7	≈ 3.51
CosPlace + SuperGlue	76.2/80.3/82.5	72.9/75.7/76.4	87.7/92.6/94.4	72.4/76.9/78.5	≈ 1.78
CosPlace + SP+LG	76.5/80.6/81.9	73.8/76.0/76.5	89.8/93.1/94.6	73.3/76.9/79.2	≈ 3.30
MegaLoc (R@20)	98.7	91.5	99.6	99.4	
MegaLoc	95.6/97.8/98.7	85.6/89.4/90.0	97.2/99.3/99.5	97.1/98.9/99.1	-
MegaLoc + LoFTR	94.0/97.5/98.4	86.5/89.7/90.8	97.3/99.3/99.5	92.2/98.4/99.1	≈ 4.06
MegaLoc + SuperGlue	92.1/97.1/98.4	85.8/89.9/90.7	93.8/98.7/99.4	90.2/98.1/98.9	≈ 1.71
MegaLoc + SP+LG	93.0/97.5/98.1	87.0/90.6/91.3	96.0/99.1/99.4	91.10/97.9/99.0	≈ 3.74
MixVPR (R@20)	93.7	83.9	95.9	88.0	
MixVPR	78.1/89.5/92.4	70.2/79.0/81.3	85.4/93.0/94.7	62.9/79.8/84.1	-
MixVPR + LoFTR	89.8/92.4/93.7	80.1/82.9/83.7	93.4/95.0/95.3	82.5/86.8/87.6	≈ 3.37
MixVPR + SuperGlue	86.7/91.8/93.3	80.1/82.7/83.2	89.8/94.0/95.2	81.9/86.4/87.5	≈ 1.50
MixVPR + SP+LG	89.2/92.4/92.7	81.4/83.3/83.7	91.7/95.0/95.6	81.9/86.2/87.1	≈ 3.13

Table 1. Baseline table for performance measurements.

ment steps.

4. Adaptive re-ranking strategy

4.1. Method Overview

The standard Visual Place Recognition pipeline applies image matching-based re-ranking to all queries indiscriminately, which can lead to significant computational overhead. This work proposes an adaptive re-ranking strategy that selectively applies geometric verification only to queries predicted as “hard”, thereby reducing computational costs while maintaining or improving performance. We define a hard query as one where the number of inliers between the query and the first retrieved database image is low relative to the distribution across all queries.

The proposed approach follows a four-stage pipeline:

1. **Inliers Analysis:** Aggregate inlier count statistics from image matching on training data (SVOX Sun and Night subsets) using two image matching algorithms (LoFTR and SuperGlue). We characterize the relationship between inlier counts and query correctness to validate that the inlier count serves as a reliable predictor of query difficulty.
2. **Logistic Regression Training:** Train dataset-specific logistic regression models that map inlier counts to a probability score indicating whether a query is “hard” and should undergo full geometric re-ranking. This probabilistic formulation avoids hard thresholding and successfully captures dataset-specific inlier distributions.
3. **Threshold Validation:** On validation data (SF-XS), sweep over a range of decision thresholds and evaluate the trade-off between performance variation and computational savings. We select optimal thresholds that balance these objectives for each matcher-dataset combina-

tion.

4. **Transfer Analysis:** Apply the trained models and optimal thresholds across all test datasets (Tokyo-XS, SF-XS, SVOX Sun and Night) to assess cross-dataset generalization, rigorously measuring both performance and computational efficiency gains.

The key insight is that the number of inliers between a query and its top-1 retrieval provides a strong signal about retrieval confidence. Queries with few inliers are uncertain and benefit from full re-ranking; conversely, queries with many inliers are confident and can skip unnecessary computation. This dataset-aware adaptive strategy yields significant computational savings while maintaining competitive recall performance across test datasets.

4.1.1. Standard Image Matching Procedure

Throughout all stages of our pipeline (training, validation, and testing), the image matching procedure remains consistent. For each query, we extract global features using the MixVPR and CosPlace ensemble, then perform local feature matching (LoFTR or SuperGlue) between the query and the top-1 retrieved database image. Geometric verification via RANSAC yields the inlier count, which serves as the primary signal for query difficulty estimation. This standardized procedure ensures consistency across all evaluation phases and enables a fair comparison of matcher-dataset combinations.

4.2. Inliers Analysis

The foundation of our adaptive strategy rests on understanding the relationship between the **correctness of the top-1 retrieval** (i.e., the ground truth label) and the number of inliers obtained from image matching.. We perform a comprehensive analysis across multiple image matching algorithms to characterize this relationship.

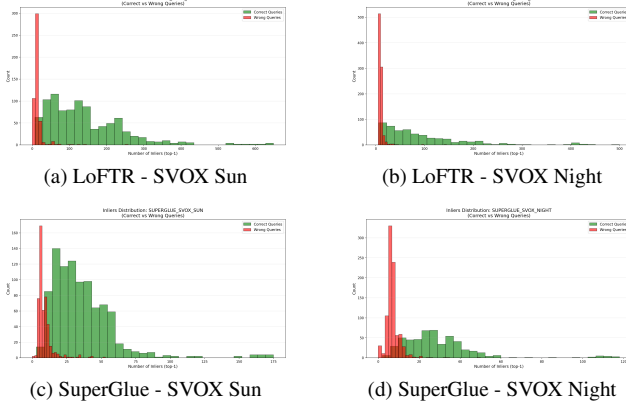


Figure 2. Inlier count distributions across all matcher-dataset combinations. The consistent bimodal separation between correct (green) and incorrect (red) queries across LoFTR and SuperGlue matchers, and across SVOX Sun and Night datasets, validates the discriminative power of inlier counts as a universal difficulty signal.

4.2.1. Data Collection and Statistics

We aggregate inlier count data from image matching operations performed on the top-1 retrieved database image for each query across the training datasets (SVOX Sun and SVOX Night subsets). For each combination of matcher and dataset, we collect pairs (x_i, y_i) where:

$$x_i = \text{num_inliers}(q_i, \text{db}_{r_i^{(1)}}) \quad (1)$$

and $y_i \in \{0, 1\}$ indicates whether the retrieval was correct (i.e., within 25 meters). Here, q_i is the i -th query image and $\text{db}_{r_i^{(1)}}$ is its top-1 retrieved database image.

For each dataset-matcher combination (SVOX Sun, SVOX Night $\times$ LoFTR, SuperGlue), we compute descriptive statistics to quantify the separation between correct and incorrect retrievals:

$$\text{Correct: } \mu_c, \sigma_c, \text{median}_c, [\min_c, \max_c] \quad (2)$$

$$\text{Incorrect: } \mu_w, \sigma_w, \text{median}_w, [\min_w, \max_w] \quad (3)$$

The analysis reveals a consistent finding: the mean inlier count for correct queries is typically 2 to 3 times higher than for incorrect queries. Figures 2 visualize these distributions across all matcher-dataset combinations. The histograms show a pronounced bimodal pattern: correct queries (green) concentrate at higher inlier counts, while incorrect queries (red) cluster at lower values. This separation is robust across both daytime and nighttime conditions (SVOX Sun vs. Night) and across both matchers, validating the inlier count as a discriminative feature for predicting query difficulty.

4.2.2. From Statistics to Predictive Modeling

The clear separation between correct and incorrect queries motivates a probabilistic modeling approach. While a fixed threshold on inlier counts might suffice for a specific matcher-dataset combination, the observed variability across datasets and matchers suggests that a dataset-specific, data-driven decision boundary is strictly preferable. A logistic regression model can learn an effective boundary, outputting a probability score $P(\text{easy} \mid x)$ that adapts to the specific characteristics of each pair. This probabilistic formulation enables a principled threshold selection based on performance-efficiency trade-offs during validation.

4.3. Logistic Regression for Query Difficulty Prediction

Building on the statistical insights from the previous section, we train logistic regression models to predict whether a query can safely skip full re-ranking based on its inlier count with the top-1 retrieved image.

4.3.1. Model Training

For each matcher-dataset combination (LoFTR and SuperGlue on SVOX Sun and SVOX Night), we train a dataset-specific logistic regression model. The model takes the input feature vector x (in our case, a scalar representing the inlier count) and outputs the probability that a query is easy:

$$P(\text{easy} \mid x) = \frac{1}{1 + e^{-(w^T x + b)}} \quad (4)$$

where b is the bias term (intercept) and w represents the learned weight for the input feature. Training follows a standard procedure: we load the aggregated inlier data, apply a stratified train-validation split (80%/20%, using random seed 42), fit the model with L2 regularization, and evaluate its performance. Matcher and Dataset-specific models are necessary because matchers have different ways to calculate inliers and inlier distributions vary significantly between daytime (SVOX Sun) and nighttime (SVOX Night) due to changing lighting conditions and visibility patterns.

4.3.2. Model Evaluation

We evaluate model performance using AUPRC (Area Under the Precision-Recall Curve), which emphasizes performance on the minority class (hard queries), and AUC-ROC (standard receiver operating characteristic). These metrics are computed on held-out validation data to strictly assess generalization before applying the models to downstream tasks.

4.3.3. Model Selection and Learned Parameters

We selected logistic regression over alternatives (e.g., SVM, Neural Networks) because: (1) our input space is one-dimensional (inlier count), and simple models prevent

overfitting; (2) the probabilistic output enables principled threshold selection; (3) learned coefficients are interpretable, enabling understanding of model behavior; (4) inference is computationally efficient. The learned weight w is positive, confirming that higher inlier counts correlate with "easy" queries. Its magnitude determines the sharpness of the easy/hard decision boundary: larger w yields steeper transitions, while the bias term b calibrates the baseline difficulty estimate.

4.4. Threshold Selection Strategy

After training dataset-specific logistic regression models, we transition from the classical binary hard-thresholding (probability > 0.5) to a principled threshold selection strategy. We employ a validation dataset to identify optimal decision thresholds that systematically balance performance gains with computational efficiency.

4.4.1. The Threshold Sweeping Process

For each matcher-dataset combination, we perform a comprehensive threshold sweeping procedure on the validation set according to the following pipeline:

1. **Extraction and Matching:** Apply the standard image matching procedure (Section 4.1.1) to extract inlier counts for all validation queries.
2. **Probability Computation:** Utilizing the trained logistic regression model, we compute the posterior probability $P(\text{easy} \mid x)$ for each query's inlier count x to determine if the query is "easy".
3. **Threshold Evaluation:** We sweep across a set of candidate thresholds $\tau \in \{0.50, 0.60, 0.70, \dots, 0.90\}$ and evaluate a composite performance score for each candidate value.

4.4.2. The Score Calculation

The objective function evaluating each candidate threshold τ combines two core metrics:

$$\text{Score}(\tau) = \alpha \cdot \text{Accuracy}_{\text{easy}}(\tau) + (1 - \alpha) \cdot \text{TimeSaved}_{\%}(\tau) \quad (5)$$

where:

- τ denotes the candidate probability threshold.
- $\alpha \in [0, 1]$ is a tunable weighting parameter governing the optimization trade-off.
- $\text{Accuracy}_{\text{easy}}(\tau)$ measures the fraction of queries predicted as "easy" (i.e., $P(\text{easy} \mid x) \geq \tau$) where the top-1 retrieved candidate is correct (within 25 meters).
- $\text{TimeSaved}_{\%}(\tau)$ represents the percentage of total computational time saved by selectively skipping the full re-ranking stage on queries classified as easy.

4.4.3. Computing Time Saved

To accurately quantify $\text{TimeSaved}_{\%}(\tau)$, we leverage empirical timing statistics harvested from initial baseline execution runs. Specifically, for each individual image matcher,

we pre-compute the average per-query processing cost, T_{query} , which encapsulates feature extraction, local feature matching, and geometric verification across 20 candidate matches.

Queries classified as easy bypass the expensive multi-matcher re-ranking stage and require only a single initial ranking verification, thereby saving the computational overhead of the remaining 19 matching operations. The total absolute computational time saved is mathematically formulated as:

$$\text{TimeSaved}(\tau) = |\mathcal{Q}_{\text{easy}}(\tau)| \times \frac{19}{20} \times T_{\text{query}} \quad (6)$$

where $|\mathcal{Q}_{\text{easy}}(\tau)|$ denotes the cardinality of the set of queries predicted as easy under threshold τ , and $\frac{19}{20}$ represents the fraction of computation time saved by skipping 19 out of 20 matching operations.

4.4.4. Optimal Threshold Selection

By dynamically varying the parameter α , we can precisely adjust the operational trade-off between maintaining a high recall rate and maximizing computational throughput. The optimal operational threshold τ^* is selected as the argument that maximizes the objective score for each distinct matcher-dataset pair:

$$\tau^* = \arg \max_{\tau} \text{Score}(\tau) \quad (7)$$

In our experiments, we set $\alpha = 0.9$, prioritizing retrieval accuracy over computational savings. This choice yields dataset-specific optimal thresholds that vary across matcher-dataset pairs. We sweep over candidate thresholds $\tau \in \{0.50, 0.60, 0.70, 0.75, 0.80, 0.85, 0.90\}$ and empirically observe optimal values $\tau^* \in \{0.50, 0.70, 0.75, 0.90\}$ across different configurations. A sensitivity analysis using $\alpha = 0.6$ (equal weighting between accuracy and speed) reveals an interesting phenomenon: all optimal thresholds collapse to $\tau^* = 0.5$, indicating that approximately 50% of queries can safely skip re-ranking when objectives are near equally weighted—computational savings dominate the optimization even under balanced weighting. This formulation yields a fully dataset-aware adaptive inference strategy.

4.5. Transfer Analysis: Cross-Dataset Generalization

While threshold validation on SF-XS provides optimal thresholds for each matcher-dataset combination, a critical remaining question is: *how do these dataset-specific models generalize to new test environments?* To rigorously assess generalization robustness and identify the most reliable model-dataset pairings, we perform an offline transfer analysis that systematically evaluates all trained models across all test datasets.

4.5.1. Offline Transfer Evaluation Protocol

Unlike runtime adaptive routing, our evaluation protocol treats transfer analysis as an *offline comparative study*. For each of the four trained logistic regression models (LoFTR Sun, LoFTR Night, SuperGlue Sun, SuperGlue Night) and each of the four test datasets (Tokyo-24/7, SF-XS test split, SVOX Sun test split, SVOX Night test split), we independently execute the following pipeline:

1. Retrieve the top-20 database candidates for each query using the MixVPR+CosPlace VPR ensemble.
2. Apply the standard image matching procedure (Section 4.1.1) to extract the inlier count x for the top-1 candidate.
3. Load the trained logistic regression model corresponding to the specific matcher-dataset pair (e.g., LoFTR trained on SVOX Sun) and compute $P(\text{easy} \mid x)$.
4. Apply the learned optimal threshold τ^* for that matcher-dataset pair: if $P(\text{easy} \mid x) \geq \tau^*$, classify as easy (skip full re-ranking); otherwise, classify as hard (perform full re-ranking on the remaining 19 candidates). Each matcher-dataset combination employs its own optimal threshold, selected to balance accuracy and computational efficiency according to that configuration’s specific inlier distribution characteristics.
5. Record the percentage of queries routed to the fast-path (“Easy%”) and standard recall metrics (Recall@1, Recall@5, Recall@10).

This procedure is repeated independently for all 16 combinations (4 trained models $\times$ 4 test datasets), producing a complete transfer table of performance evaluations.

4.5.2. Transfer Table Interpretation and Analysis

We evaluate transfer performance by systematically comparing models trained on one dataset against all four test datasets. Diagonal entries (same-dataset) represent oracle performance; off-diagonal entries quantify robustness to domain shifts. This framework guides practitioners in model selection: if labeled target data exists, train a dedicated model (diagonal performance); otherwise, select the pre-trained model with smallest off-diagonal drop to similar environments. Detailed results are presented in Section 5.5.

5. Results and Discussion

In this section, we evaluate the Adaptive Inference framework across four test environments (SVOX Sun/Night, Tokyo-XS, SF-XS), measuring computational savings and accuracy trade-offs under varying illumination and geographic conditions..

5.1. Experimental Setup

All evaluations use $K=20$ retrieval candidates, distance threshold $\tau=25m$, and $\alpha=0.90$ weighting prioritizing ac-

curacy. VPR models : CosPlace and MixVPR. Matchers: LoFTR (dense), SuperGlue (sparse). Datasets: SVOX Sun/Night, Tokyo-XS, SF-XS.

5.2. Statistical Inlier Distribution Analysis and Model Justification

To establish the mathematical foundation of the proposed adaptive framework, we examine the empirical distribution of spatial inliers generated on the top-1 retrieval candidate. Figures 2a, 2b, 2c and 2d, illustrate the histograms of inlier counts, while explicit statistical parameters are reported in Table 2.

All four distributions show a clear two-part separation. Incorrect matches (red bars) produce consistently low inlier counts: median of 11 for LoFTR, 6–7 for SuperGlue. This pattern holds uniformly across all datasets and matchers, because incorrect image pairs simply lack the geometric correspondence needed for feature alignment.

Conversely, geographically correct retrievals exhibit a significantly broader, high-consensus distribution, which is strongly governed by two distinct phenomena:

- **Dense vs. Sparse Discrepancy:** For the dense matcher (LoFTR), correct matches under *SVOX Sun* spread into an extensive “long-tail” distribution reaching up to 649 inliers with a high median of 121. For SuperGlue, the sparse feature matching topology bounds the maximum inlier count to 175, maintaining a tighter distribution centered around a median of 31 inliers.
- **Day-to-Night Photometric Suppression:** Comparing the daytime distributions against the nighttime counterparts reveals a severe leftward shift and compression of the true-positive distributions. Under nighttime conditions, low contrast and sensor noise sharply suppress descriptor matching density. For LoFTR, the median of the correct matches collapses from 121 down to a fragile 67.50. For SuperGlue, the nighttime true-positive median gets compressed to 26.50, resulting in a substantial distribution overlap with the false-positive peak in the 5 to 15 inlier fringe zone.

This behavior strongly supports the choice of a parametric Logistic Regression model. Because true and false hypotheses are separated by a distinct geometric boundary, the log-linear model can accurately fit a smooth sigmoid curve capable of mapping raw inlier counts into a well-calibrated posterior probability $P(y = 1 \mid x)$.

Furthermore, the significant overlap observed in the nighttime distribution mathematically explains why there is a highly stringent threshold (0.90 for LoFTR and 0.70 for SuperGlue) under the conservative $\alpha = 0.90$ regime. The model is forced to be mathematically strict because a low inlier count in the night domain is highly ambiguous, whereas a high inlier count in the day domain strongly correlates with localization success.

Dataset	Matcher	Query Type	Mean $\pm$ Std	Median	Range
svox_sun	LoFTR	Correct	142.00 $\pm$ 105.58	121.00	[7, 649]
		Wrong	14.85 $\pm$ 17.23	11.00	[0, 258]
	SuperGlue	Correct	35.93 $\pm$ 23.97	31.00	[3, 175]
		Wrong	8.83 $\pm$ 5.61	7.00	[0, 53]
svox_night	LoFTR	Correct	92.73 $\pm$ 87.18	67.50	[6, 500]
		Wrong	10.17 $\pm$ 7.29	9.00	[5, 145]
	SuperGlue	Correct	29.01 $\pm$ 18.57	26.50	[2, 118]
		Wrong	6.80 $\pm$ 3.55	6.00	[0, 50]

Table 2. Empirical Inlier Statistics for Top-1 Hypotheses Extracted via Spatial Verification

5.3. Quantitative Performance and Accuracy-Latency Trade-Off

The macroscopic results of the adaptive routing strategy for LoFTR and SuperGlue are systematically reported in Table 3 and Table 4, respectively.

The experimental results validate a massive reduction in runtime latency. In benign, well-illuminated environments like *SVOX Sun*, the adaptive selector flags a dominant portion of the queries as “Easy”, ranging from 73.2% (LoFTR) up to 78.8% (SuperGlue). For these queries, the framework performs an early exit immediately after evaluating the top-1 candidate, completely bypassing the feature extraction and matching loops across the remaining $K - 1$ gallery items. Crucially, this early-stopping mechanism introduces negligible degradation to accuracy, securing a high Recall@1 (0.90 with LoFTR). This confirms that exhaustively matching the entire candidate list is computationally redundant under stable lighting conditions, where the initial prediction from the global retriever is highly informative.

5.4. Environmental Uncertainty Awareness

The routing framework effectively mirrors the underlying micro-statistical inlier behaviors within its macroscopic decisions. When evaluated on *SVOX Night*, the severe leftward compression of the correct-match inlier distribution directly affects the gating performance: the proportion of easy queries drops sharply to 40.6% – 42.6% for LoFTR and 43.2% – 46.2% for SuperGlue.

Because the low-inlier regions are highly ambiguous, the model dynamically routes more than 53% of the queries to the comprehensive geometric re-ranking pipeline. This behavior proves that the Logistic Regression model acts as a competent estimator of matching uncertainty. When spatial matchers return an impoverished inlier count due to night artifacts, the policy safely backs down and invokes full sequence verification, keeping the Recall@1 stable around 0.72 – 0.74.

5.5. Cross-Domain and Cross-Dataset Transferability

The proposed framework demonstrates powerful robustness when subjected to severe cross-condition and zero-shot geographic transfers.

5.5.1. Cross-Illumination Transferability

As evidenced in Table 3, models optimized on nighttime distributions (*SVOX Night*) maintain exceptional prediction capacity when transferred directly to a daytime environment (*SVOX Sun*), achieving 76.40% easy queries and 0.89 Recall@1. This directly correlates with our micro-statistical analysis: nighttime training forces the logistic model to establish a conservative probability threshold (0.90 for LoFTR) due to high overlap in the lower inlier spectrum.

When exposed to daytime imagery, correct pairs yield up to 649 inliers (with a median of 121). These high values instantly saturate the nighttime sigmoid curve, allowing the framework to effortlessly convert daytime feature clarity into significant computational savings.

5.5.2. Zero-Shot Geographic Transferability

When evaluated on geographically remote urban landscapes like *Tokyo* and *SF-XS*, which introduce unmapped architectural styles and large viewpoint variations, the framework generalizes seamlessly. The selector safely isolates a substantial fraction of easy queries (up to 63.3% for LoFTR on *SF-XS*), while keeping accuracy bounds remarkably stable (0.76 Recall@1 on *Tokyo* for SuperGlue across both training configurations). This demonstrates that predicting the correct matching likelihood via local keypoint inlier counts remains robust across heterogeneous environments.

5.6. Architectural Trade-Off: Dense vs. Sparse Verification

Comparing the two matching philosophies highlights an explicit architectural trade-off rooted in their feature topology.

SuperGlue demonstrates superior latency reduction and flexibility; its sparse matching formulation allows a looser confidence threshold (0.50 on *Sun*), facilitating a higher early-exit percentage (78.8%) and more aggressive computational pruning. Its transfer performance across configurations is almost invariant, maintaining an identical 0.75 Recall@1 on the challenging *SF-XS* benchmark.

Conversely, LoFTR operates with a strictly conservative thresholding policy, filtering candidates with heightened geometric severity. While this filters fewer queries as easy under extreme transfer conditions, it rewards the system on absolute precision, yielding an absolute +4.0% margin in

Training Source	Test Dataset	Total Queries	Easy %	Recall@1	Recall@5	Recall@10
SVOX SUN (Threshold: 0.75)	svox_sun	500	73.20	0.90	0.93	0.94
	svox_night	500	40.60	0.74	0.78	0.80
	tokyo	315	48.90	0.76	0.81	0.82
	sf_xs	480	56.20	0.74	0.77	0.78
SVOX NIGHT (Threshold: 0.90)	svox_sun	500	76.40	0.89	0.93	0.94
	svox_night	500	42.60	0.74	0.78	0.80
	tokyo	315	52.10	0.75	0.80	0.82
	sf_xs	480	63.30	0.70	0.75	0.78

Table 3. Quantitative Performance and Cross-Dataset Generalization for LoFTR

Training Source	Test Dataset	Total Queries	Easy %	Recall@1	Recall@5	Recall@10
SVOX SUN (Threshold: 0.50)	svox_sun	500	78.80	0.86	0.93	0.94
	svox_night	500	46.20	0.72	0.78	0.79
	tokyo	315	42.90	0.76	0.80	0.83
	sf_xs	480	46.50	0.75	0.79	0.79
SVOX NIGHT (Threshold: 0.70)	svox_sun	500	76.60	0.86	0.93	0.94
	svox_night	500	43.20	0.73	0.78	0.79
	tokyo	315	39.40	0.76	0.80	0.83
	sf_xs	480	42.90	0.75	0.79	0.79

Table 4. Quantitative Performance and Cross-Dataset Generalization for SuperGlue

Recall@1 over SuperGlue within the *SVOX Sun* landscape (0.90 vs. 0.86) due to the granular confidence provided by its high-density pixel matching.

6. Conclusion and Future works

In conclusion, this work presents a novel, adaptive re-ranking strategy that effectively mitigates the performance-efficiency trade-off in Visual Place Recognition (VPR) pipelines. By leveraging the geometric inlier count from the top-1 retrieved candidate as a reliable signal of retrieval confidence, we developed a lightweight logistic regression framework that dynamically classifies query difficulty. Extensive empirical evaluations demonstrate that this selective routing safely bypasses exhaustive geometric verification for confident queries, achieving substantial computational savings (up to 78%) while preserving high localization accuracy. Furthermore, the approach proves highly robust against severe illumination changes and zero-shot geographic domain shifts. Future works will investigate the integration of richer confidence metrics, such as the spatial distribution of inliers, to further refine the difficulty estimation. Additionally, extending this adaptive paradigm to incorporate hyper-efficient matchers like LightGlue, and exploring end-to-end uncertainty prediction directly within the global retrieval architecture, represent promising directions for scalable, real-time VPR systems.

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